

# numpy practice session

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```
In [2]: # pip instal numpy
```

```
In [4]: # import the library in j.notebook
import numpy as np
```

## create an array using numpy

```
In [5]: np.array(["Pakora", "Samosa", "Raita"])
```

```
Out[5]: array(['Pakora', 'Samosa', 'Raita'], dtype='<U6')
```

```
In [6]: price = np.array([5, 5, 5])
price
```

```
Out[6]: array([5, 5, 5])
```

```
In [8]: type(price)
```

```
Out[8]: numpy.ndarray
```

```
In [10]: len(price)
```

```
Out[10]: 3
```

```
In [14]: food = np.array(["Pakora", "Samosa", "Raita"])
```

```
In [16]: type(food)
```

```
Out[16]: numpy.ndarray
```

```
In [19]: np.ones(5)
```

```
Out[19]: array([1., 1., 1., 1., 1.])
```

```
In [22]: np.arange(10)
```

```
Out[22]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [23]: np.arange(2, 20, 2)
```

```
Out[23]: array([ 2,  4,  6,  8, 10, 12, 14, 16, 18])
```

```
In [28]: np.arange(5, 55, 5)
```

```
Out[28]: array([ 5, 10, 15, 20, 25, 30, 35, 40, 45, 50])
```

```
In [27]: np.linspace(1, 10, num=5)
```

```
Out[27]: array([ 1. ,  3.25,  5.5 ,  7.75, 10.  ])
```

```
In [31]: np.ones(50, dtype=np.int8)
```

```
Out[31]: array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1], dtype=int8)
```

```
In [32]: np.ones(50, dtype=np.int64)
```

```
Out[32]: array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1, 1], dtype=int64)
```

```
In [33]: np.ones(50, dtype=np.float64)
```

```
Out[33]: array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,  
                1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,  
                1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
```

# Array Function

```
In [36]: a = np.array([10,12,15,2,4,6,100,0.5, 320,10.3])
          a
```

```
Out[36]: array([ 10. , 12. , 15. ,  2. ,  4. ,  6. , 100. ,  0.5, 320. ,
                  10.3])
```

```
In [37]: a.sort()  
a
```

```
Out[37]: array([ 0.5,  2. ,  4. ,  6. , 10. , 10.3, 12. , 15. , 100. ,
                320. ])
```

```
In [38]: b = np.array([10,3,24,65,3.5,345])
          b
```

```
Out[38]: array([ 10. ,   3. ,  24. ,  65. ,   3.5, 345. ])
```

```
In [39]: c = np.concatenate((a,b))
```

```
Out[39]: array([ 0.5,  2. ,  4. ,  6. , 10. , 10.3, 12. , 15. , 100. ,
                320. , 10. ,  3. , 24. , 65. ,  3.5, 345. ])
```

```
In [40]: c.sort()
c
```

```
Out[40]: array([ 0.5,  2. ,  3. ,  3.5,  4. ,  6. , 10. , 10. , 10.3,
                12. , 15. , 24. , 65. , 100. , 320. , 345. ])
```

## 2 D array

```
In [51]: a = np.array([[1,2,3,4],
                       [5,6,7,8]])
a
```

```
Out[51]: array([[1, 2, 3, 4],
                [5, 6, 7, 8]])
```

```
In [52]: b = np.array([[6,7, 9, 11],[8,9, 10, 12]])
b
```

```
Out[52]: array([[ 6,  7,  9, 11],
                [ 8,  9, 10, 12]])
```

```
In [55]: c = np.concatenate((a,b), axis=0)
c
```

```
Out[55]: array([[ 1,  2,  3,  4],
                [ 5,  6,  7,  8],
                [ 6,  7,  9, 11],
                [ 8,  9, 10, 12]])
```

```
In [56]: c = np.concatenate((a,b), axis=1)
c
```

```
Out[56]: array([[ 1,  2,  3,  4,  6,  7,  9, 11],
                [ 5,  6,  7,  8,  8,  9, 10, 12]])
```

```
In [61]: import numpy as np

a = np.array([[[0, 1, 2, 3],
               [4, 5, 6, 7]],
              [[0, 1, 2, 3],
               [4, 5, 6, 7]],
              [[0, 1, 2, 3],
               [4, 5, 6, 7]]])

print(a)
```

```
[[[0 1 2 3]
   [4 5 6 7]]

 [[0 1 2 3]
  [4 5 6 7]]

 [[0 1 2 3]
  [4 5 6 7]]]
```

```
In [62]: # to find the number of dimensions
a.ndim
```

```
Out[62]: 3
```

```
In [65]: b = np.array([[5,6,7],
                       [8,9,10],
                       [11,13,13]])
b
```

```
Out[65]: array([[ 5,  6,  7],
                [ 8,  9, 10],
                [11, 13, 13]])
```

```
In [66]: b.ndim
```

```
Out[66]: 2
```

```
In [67]: a.size
```

```
Out[67]: 24
```

```
In [68]: a.shape
```

```
Out[68]: (3, 2, 4)
```

```
In [78]: a = np.arange(9)
a
```

```
Out[78]: array([0, 1, 2, 3, 4, 5, 6, 7, 8])
```

```
In [79]: b = a.reshape(3,3)
b
```

```
Out[79]: array([[0, 1, 2],
                [3, 4, 5],
                [6, 7, 8]])
```

```
In [82]: np.reshape(a, (1, 9), order='C')

print(a)
```

```
[0 1 2 3 4 5 6 7 8]
```

```
In [87]: a = np.array([1,2,3,4,5,6,7,8,9])
a
```

```
Out[87]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [88]: a.shape
```

```
Out[88]: (9,)
```

```
In [89]: b = a[np.newaxis, :]  
b
```

```
Out[89]: array([[1, 2, 3, 4, 5, 6, 7, 8, 9]])
```

```
In [91]: b.shape
```

```
Out[91]: (1, 9)
```

```
In [90]: c = a[:, np.newaxis]  
c
```

```
Out[90]: array([[1],  
                [2],  
                [3],  
                [4],  
                [5],  
                [6],  
                [7],  
                [8],  
                [9]])
```

```
In [92]: a[2:9]
```

```
Out[92]: array([3, 4, 5, 6, 7, 8, 9])
```

```
In [93]: a*6
```

```
Out[93]: array([ 6, 12, 18, 24, 30, 36, 42, 48, 54])
```

```
In [94]: a+6
```

```
Out[94]: array([ 7,  8,  9, 10, 11, 12, 13, 14, 15])
```

```
In [96]: a.sum()
```

```
Out[96]: np.int32(45)
```

```
In [98]: a.mean()
```

```
Out[98]: np.float64(5.0)
```

```
In [102... a.max()
```

```
Out[102... np.int32(9)
```